

AJAY VENKATESH

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Skills

Languages: TypeScript, JavaScript (ES6), HTML5, CSS3, SQL, Java, Python

Frameworks & Libraries: Angular, RxJS, NgRx, React, Redux, Node.js, Express.js

Database: RDBMS (MySQL, PostgreSQL), NoSQL (DynamoDB, Firebase, MongoDB), Dataverse

Cloud: AWS (AWS CDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM), Azure (OpenAI services, Security Data lakes, Storage blobs)

Tools: Git, Visual Studio, Cursor, Postman, Docker, Kubernetes, Power BI, Jest, Jasmine, Karma, React Testing Library

Experience

CampusMesh

Jan 2026 – Present

Software Engineer

New York, NY

- Developed scalable **Node.js** **microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
- Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda**, **API Gateway**, **S3**, and **CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, WA

- Architected a **React 18** and **TypeScript** console with **OAuth 2.0** and corporate **SSO** authentication; enforced **4-tier role-based access control** governing which routes, UI panels, and actions each user could access — protecting evidence data for **500+** fulfillment team members.
- Eliminated redundant API calls by caching server responses on the client for high-traffic dashboard views; applied page-level code splitting so each route loaded only its required bundles — cutting page load speed by **40%** and improving Core Web Vitals scores.
- Maintained zero critical defects across a 3-month release cycle by writing unit and integration tests covering auth flows, role guard behavior, and data-fetching components using **Jest** and **React Testing Library**.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus)

New York, NY

- Built a **TypeScript** and **React** frontend for an **NSF-funded (\$100K)** AI health platform, integrating real-time wearable sensor data with cloud-deployed machine learning models for iOS and Android users.
- Delivered an insurance analytics platform (**React**, **Node.js**, **REST APIs**) where brokers, underwriters, and admins each accessed role-restricted dashboards scoped to their permitted data; integrated **WebSocket** real-time updates, **responsive HTML5/CSS3** layouts, and Stripe billing — generating **\$50K+** across **6 enterprise clients**.
- Built a shared **React** component library with consistent design standards, cutting UI development time across 3 product features; coordinated **Git** branching and code review workflows across a multi-developer team.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Designed and delivered custom **Power Apps Component Framework (PCF)** controls in **TypeScript**, **HTML5**, and **CSS3** for **Dynamics 365 CRM**, replacing out-of-box UI with accessible data interaction components across **3 enterprise** deployments.
- Built **Angular** and **TypeScript** frontends integrated with Microsoft CRM, handling asynchronous, event-driven data streams across health platform modules with reactive state management.
- Established frontend code review standards covering **HTML/CSS** accessibility, component design patterns, and test coverage; reduced production defects by **25%** and mentored **3 junior engineers**.

Projects

Task Management Dashboard (Angular SPA) | *Angular, TypeScript, RxJS, NgRx, HTML5/CSS3, Jest*

- Built a single-page **Angular** application with **NgRx** for centralized state management and reactive data streams; delivered drag-and-drop task boards and real-time status updates via WebSockets.
- Achieved **90%+ test coverage** with **Jest** across components, services, and state management layers; enforced strict **TypeScript** typing and type-safe selectors throughout.

Hospital Management System (Full-Stack Web Application) | *React, TypeScript, Node.js, Express, MongoDB, AWS SageMaker*

- Developed a full-stack healthcare platform with **React** and **TypeScript**, featuring responsive **HTML5/CSS3** layouts and role-based access control to manage appointments, patient records, and lab reports.
- Fine-tuned a **Hugging Face** transformer model to **88% diagnostic accuracy** and deployed it on **AWS SageMaker**, enabling real-time condition predictions within the physician dashboard.

Education

New York University | *Master of Science in Computer Engineering (GPA – 3.9/4)*

Sep 2024 – May 2026

- Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Certifications

AWS Cloud Practitioner 

Azure Data Fundamentals 

Microsoft Solution Architect Expert 

Power BI Data Analyst Associate 

Microsoft Developer Associate 

Stanford Machine Learning (Coursera) 